

Auteur: Luca Letroye  
 Studierichting: Informatica  
 Oefeningenreeks 4A nr. 45.2

$$\begin{aligned} \int \frac{18x^2}{x^4 - 4x + 4} dx &= \int \frac{18x^2}{(x-1)^2(x^2 + 2x + 3)} dx \\ \frac{18x^2}{(x-1)^2(x^2 + 2x + 3)} &= \frac{A}{(x-1)^2} + \frac{B}{(x-1)} + \frac{Cx + D}{x^2 + 2x + 3} \\ A &= \frac{18x^2}{x^2 + 2x + 3} \Big|_{x=1} = 3 \\ \frac{18x^2}{(x-1)^2(x^2 + 2x + 3)} - \frac{3}{(x-1)^2} &= \frac{3(5x + 3)}{(x-1)(x^2 + 2x + 3)} \\ \Rightarrow B &= \frac{3(5x + 3)}{x^2 + 2x + 3} \Big|_{x=1} = 4 \\ 18x^2 &= 3(x^2 + 2x + 3) + B(x-1)(x^2 + 2x + 3)(Cx + D)(x-1)^2 \\ \Rightarrow 0x^3 + 4x^3 + Cx^3 &\Rightarrow C = -4 \\ x = 0 &\Rightarrow D = 3 \\ \Rightarrow 3 \int \frac{dx}{(x-1)^2} + 4 \int \frac{dx}{x-1} + \int \frac{-4x + 3}{x^2 + 2x + 3} dx \\ &= 3 \int t^{-2} dt + 4 \ln|x-1| - \int \frac{2(2x+2)}{x^2 + 2x + 3} + \int \frac{7}{x^2 + 2x + 3} dx \\ \int \frac{7}{x^2 + 2x + 3} dx &= 7 \int \frac{1}{(x+1)^2 + (\sqrt{2})^2} = \frac{7}{\sqrt{2}} \text{Bgtan} \left( \frac{x+1}{\sqrt{2}} \right) + C \\ \Rightarrow \frac{-3}{x-1} + 4 \ln|x-1| - 2 \ln|x^2 + 2x + 3| + \frac{7}{\sqrt{2}} \text{Bgtan} \left( \frac{x+1}{\sqrt{2}} \right) &+ C \end{aligned}$$

## References